Rank-27 Vivify Clean-Build Control Correction

Tactix Development Project

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Executive summary

This checkpoint corrects the performance authority for the promoted Vivify arena-order representation repair. The repair itself remains semantically exact and positive. However, the predecessor used by several initial whole-run comparisons was later found to contain compiled tracing controls and was not build-equivalent to the binary that produced the accepted 399.152-second Rank-27 result. Those comparisons remain useful for semantic evidence but are excluded from performance authority.

A fresh archive build of commit `a4b11964` reproduces the historical predecessor binary, SHA-256 `60fe8eae742d185e96aac6cd6a94c49bccc3bf23a852464f729e63e2a16ef366`, bit for bit. Clean candidate SHA-256 is `067f7f3a0566cce6a52c90d78228b51cc7444adadb464402d5339f07ed5070bd`. Direct clean controls preserve the complete normalized non-time image and show the candidate non-regressive on node017 and faster on node018. Most importantly, the exact unchanged predecessor ran in 553.948 seconds on node017 and 394.107 seconds on node018. Thus "recovering" the approximately 399-second behavior required no solver change: it was recovered by running the same executable on node018. The 159.841-second same-binary spread assigns the large absolute shift to execution environment or load, not source history.

The corrected clean frozen B28+C5 gate passes all 66 arms with every model or proof independently checked, all 33 paired logical images exact, no solve loss, and zero certificate residue. Clean candidate/predecessor PAR-2 is 1542.691/1576.572 seconds (0.978510). The Vivify repair is therefore promoted on corrected authority. Absolute Rank-27 runtime remains environmentally sensitive and the 300-second milestone remains open.

1 Identity audit and correction

Historical checked job `1646031` used exact binary `60fe8e...` and returned UNSAT in 399.152 seconds. Its frozen manifest records source base `8001b7f...` and source-diff SHA-256 `3667828ec6669c3ae5c0dd5a4b8f3e71b469a429a8dddf3e6294dd8846882e83`. Commit `a4b11964` contains exactly that production diff. Building an archive of that commit with the Release preset reproduced the historical 1,260,152-byte binary bit for bit. This rules out an uncommitted pre-promotion source state as the origin of the 399-second result.

By contrast, later predecessor SHA `7c5a38...` contains the conflict-stop and Sweep-time diagnostic controls and their associated symbols. Neither `60fe8e...` nor clean candidate `067f7f...` contains those diagnostics. Accordingly, report `report-to-andrew/2026-08-25-0825-rank27-vivify-radix-order/` is a frozen historical artifact superseded here only on its whole-run build-control claim. Its exact-stage, event-trace, checked-proof, and Claude-owned fuzz evidence remains valid.

2 Corrected Rank-27 controls

Job `1647411` ran the two exact clean binaries serially in one node017 allocation under the immutable two-CPU, 16 GiB, 1,200-second contract. Both returned exact-state UNSAT at 99% CPU, with the same complete normalized non-time SHA-256 `14f7da2d94153a2afd1776ecab8e7e49e5c2c0613dc26c43714471b7ecbe6194` and zero residue.

node017 metric	Predecessor	Candidate	Candidate delta
Internal time, s	553.948	551.564	−2.384
Vivify time, s	68.196	60.155	−8.041
Inprocessing time, s	198.858	192.124	−6.734
Outside-inprocessing time, s	355.089	359.440	+4.351
Peak RSS, KiB	631380	631468	+88

The total ratio is 0.995696 and the intended Vivify owner ratio is 0.882089. This is the admissible same-allocation clean comparison: the representation repair preserves all logical effects while materially reducing its local cost.

On explicit user direction, causal job `1647447` then ran the same exact `60fe8e...` predecessor on node018. It returned exact-state UNSAT in 394.107 seconds internal and 394.59 seconds wall, recovering the historical behavior. This node-constrained measurement is not used for benchmark scoring; it establishes that the earlier 553.948-second result was not a source regression. Immediate companion job `1647466` ran clean `067f7f...` on node018 and returned the identical non-time image in 386.665 seconds internal and 387.17 seconds wall.

node018 metric	Predecessor	Candidate	Candidate delta
Internal time, s	394.107	386.665	−7.442
Vivify time, s	53.908	47.798	−6.110
Inprocessing time, s	160.522	153.326	−7.196
Outside-inprocessing time, s	233.586	233.339	−0.247
Peak RSS, KiB	631428	631500	+72

Candidate ratios are 0.981117 total and 0.886659 in Vivify. These two node018 jobs are causal checks rather than scoring evidence, but they show that the clean candidate preserves and improves the recovered regime. Taken with the node017 serial comparison, the measured facts are: logical behavior is exact, Vivify cost falls on both nodes, and no clean controlled comparison shows a total regression. The inference from the identical predecessor's 159.841-second node spread is that current machine conditions dominate the large absolute runtime variation.

3 Corrected frozen guard and correctness evidence

Campaign `benchmarks/runs/20260825-0930-BC-vivify-clean-guard/`, array `1647483` and aggregate `1647536`, replaces the confounded earlier total-performance gate. It pins exact clean predecessor `60fe8e...` and candidate `067f7f...` under the pre-Phase-4.5 resource contract. All 66/66 arms are valid: each arm has 17 independently checked SAT models and 16 independently checked UNSAT proofs; all 33 paired normalized non-time images are exact; invalidity, solve loss, cleanup failure, and certificate residue are zero. Candidate/predecessor PAR-2 is 1542.691/1576.572 seconds, ratio 0.978510. Aggregate summary SHA-256 is `440ba892c30fca94a374ebac2933dd3f92b5080216cc2b6adcb974a7eea53baa`.

The surviving semantic chain remains unchanged. Exact stage `1646181` lowers Vivify from 2.106 to 1.865 seconds with every non-time line exact. Event trace `1646183` preserves all 64 rows byte for

byte. Checked Rank-27 proof job `1646346` verifies UNSAT with pinned `drat-trim` and deletes all certificate/checker scratch. Claude Code's authoritative fuzz report for `1646352` is PASS over the frozen corpus and 10,000 random cases, including 4,966 checked SAT models, 3,451 checked UNSAT proofs, and direct changed-Vivify reach, with zero failure or residue. No frozen reference solver was rerun.

The final clean Release suite passes 73/74. Every executable, solver, proof/model, trace, cleanup, proof-rule/strategy-boundary, and solve-reset test passes. Only the known documentation-size policy fails because `PROGRESS.md` and `docs/implementation-notes.md` exceed their line budgets; this is documentation debt, not a solver failure.

4 Disposition and next actions

The Vivify arena-order repair remains promoted. It changes only the representation used to restore the already-required stable logical arena order; candidate order, exact ticks/costs, scheduling, proof effects, learning, pruning, retained state, N1–N4, and the proof-rule/strategy boundary remain unchanged. All prior Sweep, directed-bridge, BVE antecedent-admission, and Congruence closures remain closed.

The approximately 399-second behavior was recovered by placement, not by a new solver edit, so it is not a stable performance guarantee. Rank 27 remains above the 300-second milestone and convergence continues. The next owner is the preserved residual-BVE flat-slice representation prototype, which already has positive exact-state stage and clean frozen-gate evidence but remains outside the active production source. After independently completing its Rank-27 and Claude-owned fuzz requirements, resume broad search-quality and throughput localization against current instrumented Kissat.